bitcoin for doing so, which is why they choose to compete in the first place.

Competition for a financial reward is also what keeps Bitcoin's blockchain secure. If any ill-motivated actors wanted to change Bitcoin's blockchain, they would need to compete with all the other miners distributed globally who have in total invested hundreds of millions of dollars into the machinery necessary to perform PoW. The miners compete by searching for the solution to a cryptographic puzzle that will allow them to add a block of transactions to Bitcoin's blockchain.

The solution to this cryptographic puzzle involves combining four variables: the time, a summary of the proposed transactions, the identity of the previous block, and a variable called the *nonce*.

The nonce is a random number that when combined with the other three variables via what is called a cryptographic hash function results in an output that fits a difficult criteria. The difficulty of meeting this criteria is defined by a parameter that is adjusted dynamically so that one miner finds a solution to this mathematical puzzle roughly every 10 minutes. If all of this seems like drinking water out of a fire hose, that's okay—it's that way for everyone at the outset. We'll cover this process in greater detail in Chapter 4, and then go even deeper in Chapter 14.

The most important part of the PoW process is that one of the four variables is the identity of the previous block, which includes when that block was created, its set of transactions, the identity of the block before that, and the block's nonce. If